

Data Analytics

Micro Projects - Topic 4: Bank Marketing

Introduction:

You and your team are working with a major banking institution that aims to improve the effectiveness of its direct marketing campaigns for term deposit subscriptions. Despite significant outreach efforts, the success rate of these campaigns remains relatively low, resulting in inefficient use of resources.

Your task is to develop a predictive model that can forecast whether a client will subscribe to a term deposit based on demographic, financial, and campaign-related features. The dataset provided contains detailed information about customers, such as age, job, education level, account balance, loan status, contact duration, and the outcomes of previous campaigns.

You are expected to preprocess the data, select relevant features, handle categorical variables appropriately, and train suitable classification models. Based on the insights gained, you should propose ways the bank can redesign future marketing campaigns to focus more effectively on high-potential customers and improve overall efficiency.

Tasks:

- Data Understanding and Cleanup: Describe your dataset and give an insight into the data you're working with. This should also include visualization and giving an insight into its most important features.
- Choosing a model: Choose a model for your specific problem. This should include a definition of the prediction problem. Compare different models and explain which model you have chosen and why it's the best suited one for your project.
- Evaluation: Talk about the experiences you have collected during your project. What were your biggest challenges, how did you overcome them and talk about your biggest lesson learned during the project. evaluate the trained model on unseen data.
- Creating a presentation: Your presentation should include and represent the most important aspect of the tasks named above. It should also include a brief description of the single steps that you've taken to achieve your goals.

Bonus Points:

Creative solutions such as feature engineering, the collection of additional data, or novel visualization techniques are encouraged and will positively impact your bonus points.

Additional Information about some of the attributes:

Attribute	Description
default	has credit in default?
balance	Average yearly balance in euros
housing	has housing loan?
loan	has personal loan?
day, month	last contact
duration	last contact duration, in seconds
campaign	number of contacts performed during this campaign and for this client
pdays	number of days that passed by after the client was last contacted from a previous campaign (-1 means client was not previously contacted)
previous	number of contacts performed before this campaign and for this client
poutcome	outcome of the previous marketing campaign